P. R. MADHEVAN

Firmware & SoC Systems Engineer — WLAN Silicon, Low-Power Architecture, Pre-Silicon Bring-up

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SUMMARY

Firmware engineer with six years on 802.11 WLAN silicon, working where firmware meets hardware: low-power SoC architecture, pre-silicon bring-up, and host-firmware transports over PCIe, SDIO and on-die AXI. Equally at home tracing a defect through RTL and waveforms, writing the Cortex-R4 / M4 / M52 firmware that works around it, and building the debug and automation infrastructure a bring-up runs on.

TECHNICAL SKILLS

Languages C, ARM assembly (Cortex-R4 / M4 / M52), Python, TCL, Bash

Firmware & OS FreeRTOS, bare-metal bring-up, embedded Linux kernel and device drivers, CMSIS, linker scripts and memory layout, secure boot, Yocto

Wireless 802.11 a/b/g/n/ac/ax (Wi-Fi 6 / 6E), power save (DTIM, PS-Poll, U-APSD), WPA2 / WPA3, CCMP and TKIP, Wi-Fi NAN, FTM / RTT ranging, DOCSIS, Hotspot 2.0

SoC & interconnect AXI / AHB / APB, PCIe, SDIO, I2C, SPI, UART, GPIO and pinmux, M2M DMA, PMU and clock / power domains, OTP, msgbuf ring protocols

Debug & validation JTAG and ARM ADIv5 (DP / AP, MEM-AP), OpenOCD, GDB, Lauterbach TRACE32, Veloce hardware emulation, FPGA, RTL and waveform analysis, Wireshark

Automation Jenkins CI, Git, Bitbucket, Makefiles, custom static and dynamic memory-analysis tooling

EXPERIENCE

embedUR Systems — Chennai, India **May 2020 – Present**

Technical Lead (2025–present) · Senior Software Engineer (2023–2025) · Software Engineer (2020–2023)

Windows WLAN Driver — client: Infineon *Jul 2026 – Present*

- Building the DHD-based driver applications and control utility for a **Windows WLAN driver**.

WLAN Firmware & SoC Bring-up — client: Synaptics *Jun 2022 – Jul 2026*

- Designed the **MCU-WLAN sleep protocol** for a combined MCU + Wi-Fi SoC so either core can sleep independently with **zero packet loss** across full-retention, memory-retention and standby modes, keeping the MAC receiving while the MCU sleeps. Cortex-R4 and Cortex-M52 firmware over an on-die AXI link.
- **Named inventor on an invention disclosure** in 802.11 power save, covering retrieval of AP-buffered packets under high host-wake latency while preserving firmware offload.
- Eliminated spurious interrupts across an MCU-WLAN interrupt link by root-causing them to a clock-domain-crossing race in the synchroniser's deassertion path, then deferring the clear-pending step so level-triggered semantics separate genuine edges from spurious ones. CMSIS, NVIC.
- Kept a memory-to-memory DMA engine fully usable around a **hardware transfer restriction**, by proposing a runtime classifier that reroutes only the unsafe transfers through memcpy; implemented with another engineer.
- Proved out a host-firmware data path constrained to a **single DMA channel**, by building the proof of concept for re-hosting a PCIe-style msgbuf ring protocol over an on-die AXI link; the team carried it to production.
- Enabled roughly **90% of SoC development to proceed before first silicon**, by repurposing an on-die Bluetooth Cortex-M4 as the MCU host on existing dongle silicon and writing the startup assembly, FreeRTOS port and transport driver from scratch.
- Gave the program hardware debug access on emulation by bringing up **JTAG and OpenOCD on a Veloce emulator**, root-causing several RTL-level issues in the debug path and proposing the fixes. ARM ADIv5, RTL and waveform analysis.
- Let **existing register-level validation scripts run unchanged** on an SoC with no conventional host or debug transport, by extending the verification environment to reach the chip over OpenOCD; adopted by the validation teams and later reused on FPGA.
- Drove emulator bring-up of a new SoC **from reset to ping**, and contributed the architectural analysis behind a **2x transport throughput** improvement and a shorter MCU boot.
- Cut active power **~70%** by proposing and implementing a DTIM-based power-save scheme, and shipped validation releases for three WLAN chipsets on time for a customer demo by porting a RAM-bank turn-off feature, hitting a tight memory target with a three-pass linker relocation of packet pools.
- Restored throughput on stalled SDIO links by diagnosing a bidirectional-traffic stall as an RX pool-starvation loop from its fill and idle-pool counter signature, then reserving a floor of packet pools exclusively for RX DMA refill so the release side always progresses.
- Hardened WPA2 cipher handling without breaking mixed legacy networks, by restricting TKIP to the group cipher across the supplicant, host driver and firmware MAC key-install code. WPA2, CCMP.

- Sped a hardware emulator **~400%**, compressing a ROM tapeout cycle to **one week** and lifting emulator utilisation to near 100%, by rebuilding WLAN test automation and extending it into a nightly regression suite across bands, channels and security modes. Python, TCL, Jenkins CI.
- Wrote the debug tooling used through pre-tapeout bring-up: a memory analyser reconstructing system state from raw dumps (Rabin-Karp scan, branch-opcode stack unwinder), a dynamic heap profiler with flame charts, a static struct padding and cache-line analyser, a firmware preload utility, and a strap-pin automation kit that made fully remote CI reflash possible.
- Built a disassembly scan for illegal undefined-instruction patterns that surfaced **latent crash points** across the firmware and proposed the fix; separately restored system discovery by correcting a ROM-table parser inconsistency exposed once a second core was added.
- Resolved a **day-0 field escalation** by programming internal interrupt generators and an on-chip JTAG master to reach otherwise inaccessible debug registers.
- Mentored **two engineers** through bring-up and low-level debug work; both went on to earn customer appreciation for their contributions.

Embedded Linux & Industrial Switching — client: Hirschmann (Belden) *May 2021 – Jun 2022*

- Cut boot time on industrial switch platforms **from 165-172 s to 40-45 s** during a VxWorks-to-Linux migration, by tracing the delay to a 10-second retry loop in the hardware-scan path and moving module initialisation into two dedicated tasks communicating over IPC, mirroring the VxWorks design. **Pat-on-the-back award from the customer and manager.**
- Fixed module-load failures at boot by establishing that the kernel-side BDE module was not loaded ahead of its user-space counterpart — **the customer cited the depth of the analysis**. Linux kernel, device drivers.
- Sped module loading with a new algorithm supporting the platform's pluggable module cards. **Spot award.**
- Contributed to hot-pluggable SFP and PSU module support on the GRS1040 and MSP40 platforms — a Linux I2C EEPROM driver written against the vendor I2C adapter to read module parameters through a two-stage CPLD multiplexer, plus device-tree changes, a temperature-sensor race fix and a PCIe enumeration-order correction — supporting the engineer leading that work.

Cable Modem Firmware & Lab Provisioning — client: CommScope (ARRIS) *May 2020 – May 2021*

- Debugged DOCSIS signal-strength, DHCP, Green Ethernet and connection defects on SURFboard G34 gateways, across the SNMP and dmcli backends. RDKB, Yocto.
- Cut client test-machine cost **~90%** during a hardware shortage by building a Raspberry Pi replacement for the provisioning client, and brought up the lab provisioning system and CMTS server behind it.

EDUCATION

B.E. Electrical & Electronics Engineering — SRM Easwari Engineering College, Anna University, Chennai *2016 – 2020*

- **Gold Medalist, summa cum laude.**
- Higher Secondary **94%** · Secondary **93%** — Sri Sankara Vidyashramam Matriculation Higher Secondary School, Chennai.
- Final-year project: **mobile communication over TV White Space (TVWS)** — drove transmit and receive directly from Raspberry Pi GPIO pins instead of dedicated RF transceiver chips, with Reed-Solomon forward error correction and POCSAG signalling.

RECOGNITION

- **Named inventor on a filed invention disclosure** in 802.11 power save (AP-buffered packet retrieval under host-wake latency).
- **Customer award** for the emulator automation work · **pat-on-the-back award** from the customer and manager for the switch boot-time reduction · **spot award** for the pluggable-module loading algorithm · multiple customer recognition awards.
- Rated **5 / 5, "significantly exceeded job role"**, for three consecutive annual review cycles.